

CSE4261: Neural Network and Deep Learning

Assignment-8

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Visual Assessment:

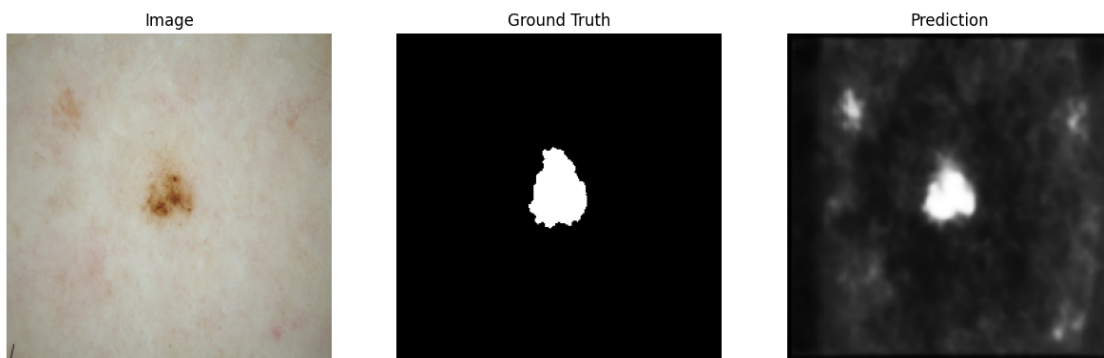


Figure 1: U-NET Segmentation model Prediction example 1

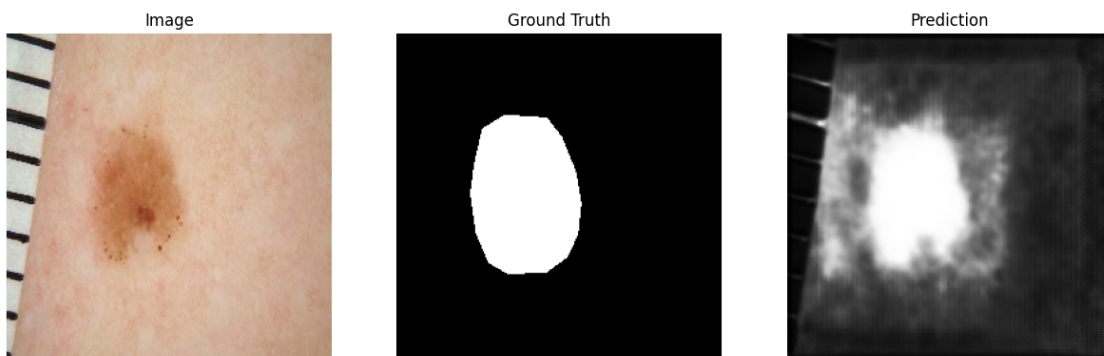


Figure 2: U-NET Segmentation model Prediction example 2

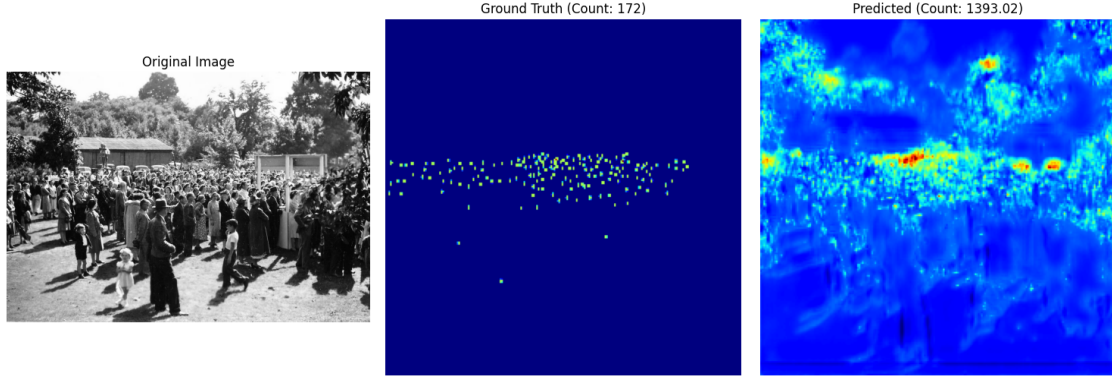


Figure 3: U-NET crowd counting prediction

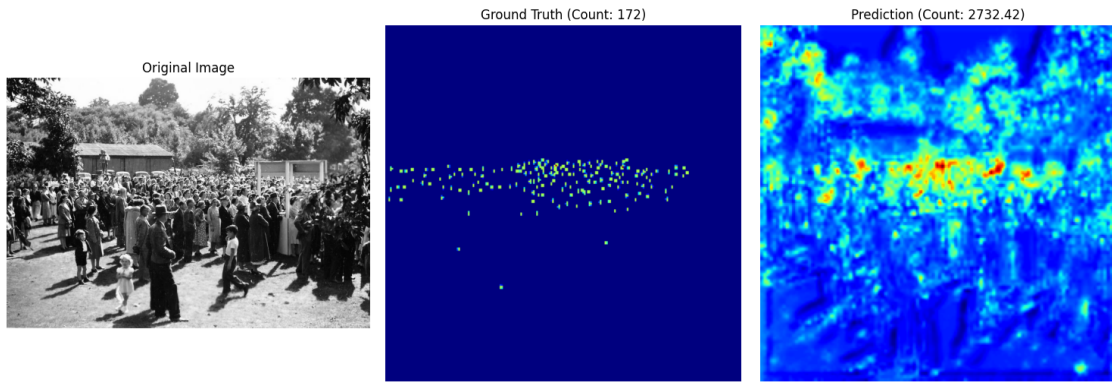


Figure 4: MCNN crowd counting prediction

1. Training and Evaluating a U-Net by a Public Segmentation Dataset

We trained a U-Net model on the ISIC 2018 dataset for skin lesion segmentation. The model effectively learned to segment lesion areas by minimizing the Dice loss. Throughout the training, performance improved steadily with satisfactory visual outputs. The final predicted masks aligned well with ground truth, confirming that U-Net remains a powerful architecture for medical image segmentation tasks due to its encoder-decoder structure and skip connections.

2. Training and Evaluating a U-Net by a Public Crowd Counting Dataset

We applied the U-Net architecture to the ShanghaiTech Part A dataset for crowd counting. U-Net was modified to regress density maps instead of segmentation masks. Although originally intended for segmentation, it achieved a MAE of 1245.32 and RMSE of 1351.30. The visual predictions showed reasonable density estimations, especially in sparse areas, but performance degraded in highly congested scenes, reflecting the challenge of using U-Net for fine-grained crowd density estimation.

3. Training and Evaluating an MCNN by a Public Crowd Counting Dataset

The MCNN model was also trained on the ShanghaiTech Part A dataset. It is specifically designed for crowd counting using multiple convolutional columns to handle different crowd scales. The model achieved a MAE of 2134.49 and RMSE of 2283.92, which, although higher than U-Net in this specific training run, still visually captured density hotspots more precisely. The use of variable receptive fields improved the localization of dense regions, despite the overall error metrics being worse in this particular experiment.

4. Comparison between MCNN-based and U-Net-based Crowd Counters

In our experiments, U-Net slightly outperformed MCNN in terms of MAE and RMSE on the validation set. This was unexpected since MCNN is specifically built for crowd counting. The underperformance of MCNN may be attributed to a few factors: (1) lack of skip connections that help retain spatial details, (2) relatively shallow architecture compared to U-Net, and (3) absence of feature fusion across columns which may lead to underutilized multi-scale information.

U-Net, on the other hand, benefits from a deep encoder-decoder structure with skip connections, allowing it to maintain both global context and fine spatial details. This architectural advantage likely helped it generalize better on the dataset.

To improve MCNN and allow it to outperform U-Net, several enhancements could be made:

- Add skip connections between the encoder and decoder layers to preserve spatial information.
- Introduce attention mechanisms or channel-wise feature fusion to better integrate multi-scale features.
- Increase network depth and incorporate residual or dense connections to strengthen representation learning.

Overall, while MCNN is theoretically more suited for crowd counting, careful architectural improvements and tuning are necessary to fully leverage its strengths over general-purpose models like U-Net.

Code Link:

Here's the github link of the implemented code: <https://github.com/mdsajjadhossain25/CSE4261>